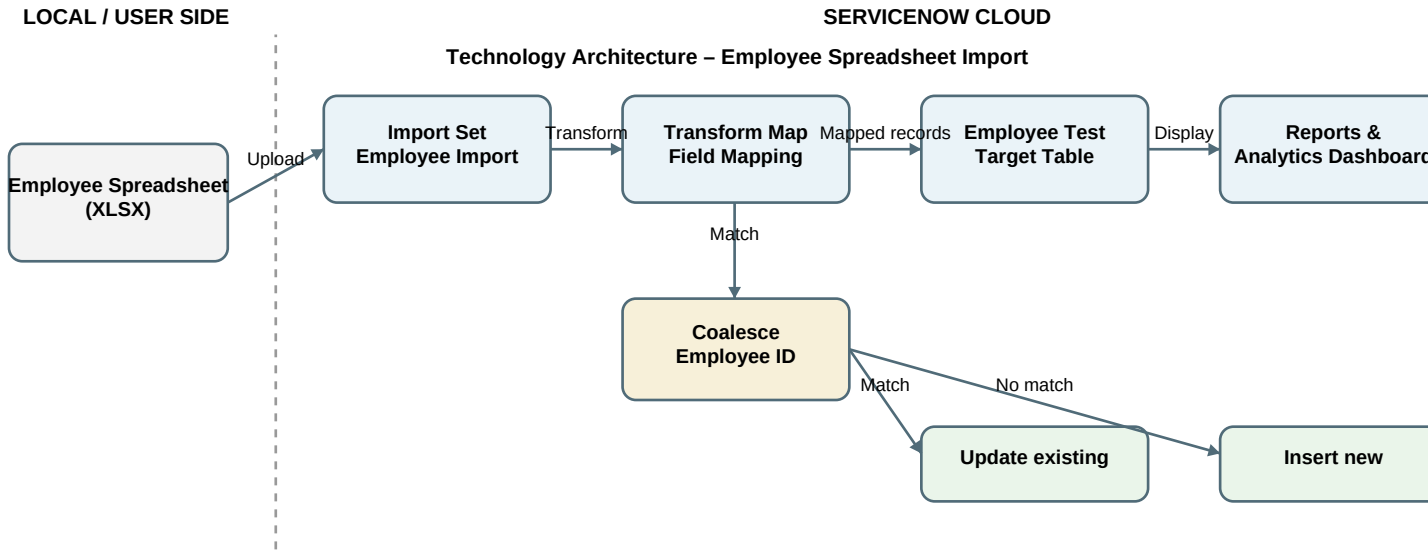
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-9088
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture

The architecture represents the project as a ServiceNow-based spreadsheet import workflow. The employee spreadsheet is the source input, ServiceNow Import Sets provide the staging area, the Transform Map performs field mapping and transformation, the Employee Test table stores the target records, and reports/dashboard provide the output view. Coalesce uses Employee ID to support update-versus-insert behavior during repeated imports.



Architecture interpretation: The spreadsheet is prepared on the local/user side and uploaded into the ServiceNow cloud environment. The remaining import, transformation, storage, reporting, and dashboard functions are handled within the ServiceNow platform.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application and project functions.	ServiceNow web UI / browser
2	Application Logic-1	Logic for loading spreadsheet data into the staging area.	ServiceNow Import Sets / Employee Import
3	Application Logic-2	Logic for mapping and transforming source data to the target table.	ServiceNow Transform Maps
4	Application Logic-3	Logic for identifying existing employee records during repeated imports.	Transform Map Field Map with Coalesce using Employee ID
5	Database	Target data storage for transformed employee records.	ServiceNow Employee Test custom table (u_employee_test)
6	Cloud Database	Platform-managed storage used by the ServiceNow instance.	ServiceNow platform-managed database
7	File Storage	Source spreadsheet file used for the import.	Local spreadsheet file / XLSX
8	External API-1	External API integration used in the application.	Not applicable
9	External API-2	External API integration used in the application.	Not applicable
10	Machine Learning Model	Machine learning interface used in the application.	Not applicable
11	Infrastructure (Server / Cloud)	Application deployment environment.	ServiceNow cloud instance; browser/local system for source spreadsheet

External interfaces and ML: The supplied project implementation describes Import Sets, Transform Maps, Coalesce, reports, and a dashboard, but does not specify third-party APIs or machine-learning models. Therefore these entries are marked Not applicable rather than adding unsupported technologies.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is identified in the supplied project implementation. The solution uses native ServiceNow platform features.	ServiceNow platform features
2	Security Implementations	Access to the application and employee data is handled through the ServiceNow platform and its configured user access controls.	ServiceNow platform access / roles as configured
3	Scalable Architecture	The solution uses ServiceNow Import Sets and Transform Maps for bulk spreadsheet processing rather than manual record-by-record creation. Repeated imports are supported through Coalesce.	ServiceNow Import Sets, Transform Maps, Coalesce
4	Availability	The solution is hosted within the ServiceNow instance; the import workflow, target table, reports, and dashboard are available when the configured ServiceNow instance is available.	ServiceNow cloud platform
5	Performance	Bulk employee data is staged and transformed through Import Sets and Transform Maps, supporting more efficient processing than manual entry of each employee record.	ServiceNow Import Sets / Transform Maps

Technology Stack Summary

The project is primarily a native ServiceNow solution. Its core stack is the ServiceNow web interface, Import Sets, Transform Maps, Coalesce, the custom Employee Test table, and ServiceNow reporting/dashboard features. The only external/local artifact explicitly required by the workflow is the employee spreadsheet used as the source data file.

Source alignment: The uploaded Technology Stack template asks for architecture, infrastructure demarcation, external interfaces, data storage, ML interfaces, component technologies, and application characteristics. The entries above are restricted to what is supported by the supplied project materials; where the project does not use a requested category, it is explicitly marked Not applicable.